SOLUTIONS: DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

1. a) Let the realisation be partitioned compatibly with P and Q as

$$G(s) \stackrel{s}{=} \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & 0 \end{array}\right] \cdot \text{ Then}$$

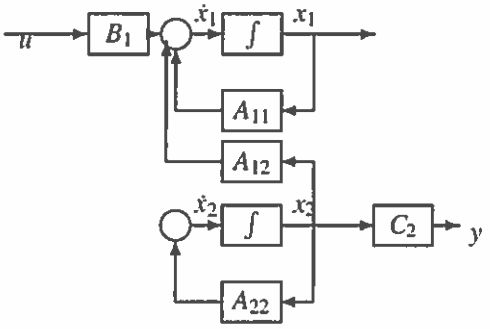
$$A^TQ+QA+C^TC = \begin{bmatrix} C_1^TC_1 & A_{21}^TQ_2+C_1^TC_2 \\ \star & A_{22}^TQ_2+Q_2A_{22}+C_2^TC_2 \end{bmatrix} = 0 \qquad (1.1)$$

$$AP+PA^T+BB^T = \begin{bmatrix} A_{11}P_1+P_1A_{11}^T+B_1B_1^T & P_1A_{21}+B_1B_2^T \\ \star & B_2B_2^T \end{bmatrix} = 0 (1.2)$$

It follows that $C_1 = 0$, $B_2 = 0$ and $A_{21} = 0$. Thus the realisation decomposes into two subsystems $G_1 : \dot{x}_1 = A_{11}x_1 + B_1u + A_{12}x_2$ with n_1 modes and $G_2 : \dot{x}_2 = A_{22}x_2$, $y = C_2x_2$ with n_2 modes. Since A is stable, both sub–systems are stable.

i) G_1 is unobservable since $C_1 = 0$. We prove controllability. Let $z'A_{11} = \lambda z'$, $z \neq 0$. It follows from the (1,1) entry of (1.2) that $(\lambda + \bar{\lambda})z'P_1z + z'B_1B_1'z = 0$. If $z'B_1 = 0$ then $\lambda + \bar{\lambda} = 0$ which contradicts the assumption that A is stable. Thus the realisation is controllable.

ii) For the subsystem G_2 it is clear that all the modes are uncontrollable and observable using a dual argument to the above.



b) i) Using the result that A is stable if and only if there exists $P = P^T \succ 0$ such that $AP + PA^T \prec 0$ we have $\mathscr{S}$ is simultaneously stabilisable if and only if there exist $K \in \mathscr{R}^{m\times n}$ and $P_i = P_i^T \succ 0$ such that $(A_i + B_iK)P_i + P_i(A_i + B_iK)^T \prec 0$, $i = 1,\ldots,N$.

ii) The above conditions are nonlinear, but can be linearised by taking $P_i = P$, and defining a variable $\hat{K} := KP$ and so a set of sufficient conditions is: $\mathscr{S}$ is simultaneously stabilisable if there exist $\hat{K} \in \mathscr{R}^{m\times n}$ and $P = P^T \succ 0$ such that $A_iP + PA_i^T + B_i\hat{K} + \hat{K}^TB_i^T \prec 0$, $i = 1,\ldots,N$. If the conditions are satisfied, then $K = \hat{K}P^{-1}$.

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2. a) i) The $(1,1)$ block of the inequality gives $A^TP+PA \prec 0$. Let $Az=\lambda z$, $z\neq 0$. Then multiplying the inequality from the left by z^T and from the right by z gives $(\lambda+\bar{\lambda})z^TPz<0$. Since $P\succ 0$ it follows that $z^TPz>0$ and it follows that $\lambda+\bar{\lambda}<0$ so that A is stable.

ii) Since A is stable, $\|H\|_\infty<\gamma$ if and only if, with $x(0)=0$, $J:=\int_0^\infty[y^Ty-\gamma^2u^Tu]dt<0$, for all $u(t)$ such that $\|u\|_2<\infty$. If $\|u\|_2$ is bounded, then $\lim_{t\to\infty}x(t)=0$. Now, $\int_0^\infty\frac{d}{dt}[x^TPx]dt=x(\infty)^TPx(\infty)-x(0)^TPx(0)=0$. So,

$$0=\int_0^\infty\left(\dot{x}^TPx+x^TP\dot{x}\right)dt=\int_0^\infty[x^T(A^TP+PA)x+x^TPBu+u^TB^TPx]dt.$$

Use $y=Cx+Du$ and add the last expression to J

$$J=\int_0^\infty[x^T(A^TP+PA+C^TC)x+2x^T(PB+C^TD)u+u^T(D^TD-\gamma^2I)u]dt$$

$$=\int_0^\infty\begin{bmatrix}x^T & u^T\end{bmatrix}\overbrace{\begin{bmatrix}A^TP+PA+C^TC & PB+C^TD\\ B^TP+D^TC & D^TD-\gamma^2I\end{bmatrix}}^{M}\begin{bmatrix}x\\ u\end{bmatrix}dt.$$

It follows that $J<0$, and so $\|H\|_\infty<\gamma$, if $M\prec 0$. This proves the result.

iii) Now, $H(s)^T=D^T+B^T(sI-A^T)^{-1}C^T$. It follows that the dual conditions are: there exists a $P=P^T\succ 0$ such that

$$\begin{bmatrix}AP+PA^T+BB^T & PC^T+BD^T\\ CP+DB^T & DD^T-\gamma^2I\end{bmatrix}\prec 0.$$

b) i) Substituting $u(t)=Kx(t)$ into the dynamic equations gives

$$\dot{x}(t)=\underbrace{(A+BK)}_{A_c}x(t)+\underbrace{B_d}_{B_c}d(t),\qquad z(t)=\underbrace{(C+DK)}_{C_c}x(t)+\underbrace{D_d}_{D_c}d(t).$$

It follows that $T_{zd}(s)=D_c+C_c(sI-A_c)^{-1}B_c$.

ii) Using the results of Part (aiii), by replacing A, B, C and D by A_c, B_c, C_c and D_c, we have that there exists a feasible K if there exists $P=P^T\succ 0$ such that

$$\begin{bmatrix}(A+BK)P+P(A+BK)^T+B_dB_d^T & P(C+DK)^T+B_dD_d^T\\ (C+DK)P+D_dB_d^T & D_dD_d^T-\gamma^2I\end{bmatrix}\prec 0.$$

Noting that the only nonlinearity is due to the product KP, we define $\hat{K}:=KP$ and so there exists a feasible K if there exists $P=P^T\succ 0$ such that

$$\begin{bmatrix}AP+PA^T+B\hat{K}+\hat{K}^TB^T+B_dB_d^T & PC^T+\hat{K}^TD^T+B_dD_d^T\\ CP+D\hat{K}+D_dB_d^T & D_dD_d^T-\gamma^2I\end{bmatrix}\prec 0.$$

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3. a) It is assumed that the triple (A,B,C) is minimal and that D has full column rank (so that D^TD is nonsingular). A suitable Lyapunov function for regulating x is $V=x^TPx$ where $P=P^T$.

b) Set $u=-Fx$. Provided that $P=P^T\succ 0$ and $\dot{V}<0$ along closed-loop trajectories, we can assume $\lim_{t\to\infty}x(t)=0$. Then $\dot{V}=\dot{x}^TPx+x^TP\dot{x}=x^T\left(A^TP+PA-F^TB^TP-PBF\right)x$. Using $x(\infty)=0$, $\int_0^\infty \dot{V}(t)dt=\int_0^\infty x^T\left(A^TP+PA-F^TB^TP-PBF\right)xdt=-x_0^TPx_0$.

c) Let $R=D^TD$. Adding the last equation to the expression for J and completing a square:

$$J=x_0^TPx_0+\int_0^\infty\{x^T[A^TP+PA+C^TC-PBR^{-1}B^TP]x+\left\|D(F-R^{-1}B^TP)x\right\|^2\}dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of F is given by $F=R^{-1}B^TP$. We can set the term in square brackets to zero provided P satisfies the Riccati equation $A^TP+PA+C^TC-PBR^{-1}B^TP=0$. It follows that the minimum value of J is $x_0^TPx_0$.

d) We prove that $A_c:=A-BR^{-1}B'P$ is stable. The Riccati equation can be written as

$$A_c'P+PA_c+C^TC+PBR^{-1}B'P=0.$$

Let $\lambda\in\mathscr{C}$ be an eigenvalue of A_c and $y\neq 0$ be the corresponding eigenvector. Pre– and post–multiplying the Riccati equation be y^* and y respectively gives $(\lambda+\bar{\lambda})y^*Py+y^*C'Cy+y^*PBR^{-1}B'Py=0$. Since $P\succ 0$ and $y\neq 0$, $y^*Py>0$, $y^*C'Cy\geq 0$ and $y^*PBR^{-1}B'Py\geq 0$. It follows that $\lambda+\bar{\lambda}\leq 0$ with $\lambda+\bar{\lambda}=0$ if and only if $B'Py=0$ and $Cy=0$. But this implies $Ay=\lambda y$ and $Cy=0$, which contradicts the observability of the pair (A,C). This proves that $\lambda+\bar{\lambda}<0$ and so the closed loop is stable.

e) Now,

$$\begin{aligned}\dot{x} &= Ax+Bu=Ax+B(d-Fx)\\ &= (A-BF)x+Bd\\ z_1 &= Cx\end{aligned}$$

it follows that $T_{z_1d}\stackrel{s}{=}(A-BF,B,C,0)$. It follows from the bounded real lemma that $||T_{z_1d}||_\infty<1$ if there exists $P=P^T\succ 0$ such that

$$\begin{bmatrix}P(A-BF)+(A-BF)^TP+C^TC & PB\\ B^TP & -I\end{bmatrix}\prec 0$$

Using a Schur complement argument, this inequality is equivalent to

$$P(A-BF)+(A-BF)^TP+C^TC+PBB^TP\prec 0.$$

However, it follows from the Riccati equation in Part c above that $P(A-BF)+(A-BF)^TP+C^TC+PBB^TP=0$. Ignoring issues relating to strict matrix inequalities, this proves that $||T_{z_1d}||_\infty\leq 1$.

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4. a) The generalized regulator formulation is given by

$$\begin{bmatrix} z(s) \\ y(s) \end{bmatrix} = P(s) \begin{bmatrix} w(s) \\ u(s) \end{bmatrix}, \; u(s) = Fy(s), P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \stackrel{s}{=} \left[\begin{array}{c|c|c} A & B_1 & B_2 \\ \hline C_1 & 0 & 0 \\ 0 & 0 & I \\ \hline I & 0 & 0 \end{array}\right].$$

b) Let $J = \|z\|_2^2 - \gamma^2 \|w\|_2^2$ denote the cost function. Let $V = x'Xx$ and set $u = Fx$. Provided that $X = X' \succ 0$ and $\dot{V} < 0$ along the closed-loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Xx + x'X\dot{x} = x'\left(A'X + XA + F'B_2'X + XB_2F\right)x + x'XB_1w_1 + w_1'B_1'Xx.$$

Integrating from 0 to ∞ and using $x(\infty) = 0$,

$$-x_0'Xx_0 = \int_0^\infty \left[x'\left(A'X + XA + F'B_2'X + XB_2F\right)x + x'XB_1w_1 + w_1'B_1'Xx\right]dt.$$

Using the definition of J, adding the last equation, and completing two squares,

$$J = x_0'Xx_0 + \int_0^\infty \left\{x'\left[A'X + XA + C_1'C_1 - X\left(B_2B_2' - \gamma^{-2}B_1B_1'\right)X\right]x + \|Zx\|^2 - \|v\|^2\right\}dt$$

where $Z = F + B_2'X$ and $v = \gamma w - \gamma^{-1}B_1'Xx$.

c) The requirement $\|H\|_\infty \le \gamma$ is equivalent to $J \le 0$ when $x_0 = 0$. Thus sufficient conditions for $J \le 0$ are $Z = 0$ and the existence of X such that

$$A'X + XA + C_1'C_1 - X\left(B_2B_2' - \gamma^{-2}B_1B_1'\right)X = 0, \qquad X = X' \succ 0.$$

Setting $Z = 0$, the control policy is $F = -B'X$. Setting $v = 0$, the worst case disturbance is $w^* = \gamma^{-2}B_1'X$. The closed-loop with $u = Fx$ and $w = w^*$ is $\dot{x} = A_cx$ where $A_c = A - \left(B_2B_2' - \gamma^{-2}B_1B_1'\right)$ and a further condition is $Re\,\lambda_i(A_c) < 0, \; \forall\, i$.

d) Using the expression for J in Part b) and the solution in Part c) with $\gamma \to \infty$, it follows that $J \le x_0^TXx_0$ and the Riccati equation becomes

$$A'X + XA + C_1'C_1 - XB_2B_2'X = 0, \qquad X = X' \succ 0.$$

Thus, as $\gamma \to \infty$, the solution of the $\mathcal{H}_\infty$ regulator converges to that of the LQR in Question 3 (with $C = C_1, B = B_2$ and $D = I$).

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